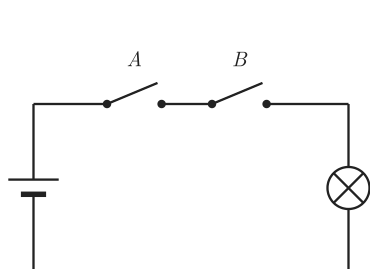


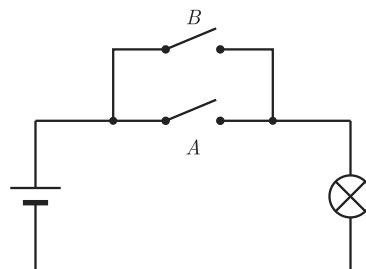
Recitation 3: Boolean algebra and logic gates

1 From switches to gates

Two switches in series with a lamp: the lamp lights only if *both* are closed. Two in parallel: it lights if *either* is closed.



both closed $\rightarrow$ lamp on (AND)



either closed $\rightarrow$ lamp on (OR)

Write 1 for closed and lamp on, 0 for open and lamp off. The two circuits become the truth tables of AND and OR:

switch <i>A</i>	switch <i>B</i>	AND series lamp	OR parallel lamp
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

The six gates from lecture:

Gate	Symbol	Algebra
NOT		$\overline{A}$
AND		$A \cdot B = AB$
OR		$A + B$
XOR		$A \oplus B$
NAND		$\overline{AB}$
NOR		$\overline{A + B}$

<i>A</i>	<i>B</i>	NOT $\overline{A}$	AND AB	OR $A + B$	XOR $A \oplus B$	NAND $\overline{AB}$	NOR $\overline{A + B}$
0	0	1	0	0	0	1	1
0	1	1	0	1	1	1	0
1	0	0	0	1	1	1	0
1	1	0	1	1	0	0	0

NAND and NOR are AND and OR with the output inverted. XOR is 1 when the inputs differ.

Notation. $\overline{A}$ and A' mean the same thing, and $\overline{AB}$ is $(AB)'$. Everything under one bar is complemented together.

Careful: $\overline{AB}$ is not $\overline{A}\overline{B}$. The first is 1 only on row 00; the second (NAND) is 1 on three rows.

Precedence. Without parentheses: NOT, then AND, then OR. So $x + yz$ means $x + (yz)$, and $\overline{xy}$ means $(\overline{x})y$, not $\overline{xy}$.

Problem 1

- Write $\overline{AB} + A\overline{B}$ with primes instead of bars. Which gate in the table is it?
- Insert all the implied parentheses in $\overline{xy} + z$.
- Are $x + yz$ and $(x + y)z$ the same function? If not, give a set of values that distinguishes them.
- Are $\overline{x} \cdot \overline{y}$ and $\overline{x \cdot y}$ the same? Try a few rows.

2 Truth tables

n inputs give 2^n rows, listed in binary counting order. Give each sub-expression its own column. Example from lecture, $A + BC + A'C'$:

A	B	C	BC	$A'C'$	$A + BC + A'C'$
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	1	0	1
1	0	0	0	0	1
1	0	1	0	0	1
1	1	0	0	0	1
1	1	1	1	0	1

A truth table proves any identity: evaluate both sides and compare the columns. It grows as 2^n , which is why we also need the algebra.

Problem 2

Fill in the output column for every input combination.

- $F = AB + B'C'$
- $G = (A + B)(A' + C)$
- How many rows would you need to check a three-variable identity? A six-variable one?

3 Boolean algebra: axioms and theorems

Axioms

- 1a** $0 \cdot 0 = 0$
1b $1 + 1 = 1$
2a $1 \cdot 1 = 1$
2b $0 + 0 = 0$
3a $0 \cdot 1 = 1 \cdot 0 = 0$
3b $1 + 0 = 0 + 1 = 1$
4a if $x = 0$, then $\bar{x} = 1$
4b if $x = 1$, then $\bar{x} = 0$

Single-variable theorems

- 5a** $x \cdot 0 = 0$
5b $x + 1 = 1$
6a $x \cdot 1 = x$
6b $x + 0 = x$
7a $x \cdot x = x$
7b $x + x = x$
8a $x \cdot \bar{x} = 0$
8b $x + \bar{x} = 1$
9 $\overline{\bar{x}} = x$

XOR identities (from lecture; Problem 3e proves two of them):

$$A \oplus 0 = A \quad A \oplus 1 = \bar{A} \quad A \oplus A = 0 \quad A \oplus \bar{A} = 1$$

Two- and three-variable theorems

- | | | | |
|--|---------------------|---|------------------|
| 10a $x \cdot y = y \cdot x$ | <i>Commutative</i> | 14a $xy + \bar{x}\bar{y} = x$ | <i>Combining</i> |
| 10b $x + y = y + x$ | | 14b $(x + y)(\bar{x} + \bar{y}) = \bar{x}\bar{y}$ | |
| 11a $x(yz) = (xy)z$ | <i>Associative</i> | 15a $\overline{xy} = \bar{x} + \bar{y}$ | <i>DeMorgan</i> |
| 11b $x + (y + z) = (x + y) + z$ | | 15b $\overline{x + y} = \bar{x} \cdot \bar{y}$ | |
| 12a $x(y + z) = xy + xz$ | <i>Distributive</i> | 16a $x + \bar{x}y = x + y$ | |
| 12b $x + yz = (x + y)(x + z)$ | | 16b $x(\bar{x} + y) = xy$ | |
| 13a $x + xy = x$ | <i>Absorption</i> | 17a $xy + yz + \bar{x}z = xy + \bar{x}z$ | <i>Consensus</i> |
| 13b $x(x + y) = x$ | | 17b $(x + y)(y + z)(\bar{x} + z) = (x + y)(\bar{x} + z)$ | |

Duality. Swap $\cdot$ with $+$, and 0 with 1, and a true identity becomes another true identity. That is why the theorems come in a/b pairs.

Problem 3

Use the tables. Name the theorem you used at each step.

- Simplify $x(x + y)$.
- Simplify $x + \bar{x}y$.
- Write the dual of the distributive law $x(y + z) = xy + xz$, and find it in the table.
- Simplify $xy + \bar{x}\bar{y} + \bar{x}y$.
- Starting from $A \oplus B = \bar{A}B + A\bar{B}$, prove $A \oplus 1 = \bar{A}$ and $A \oplus \bar{A} = 1$.

4 DeMorgan's theorem

Proof of 15a by truth table: the two right-hand columns agree in every row.

x	y	$x \cdot y$	$\overline{x \cdot y}$	$\bar{x}$	$\bar{y}$	$\bar{x} + \bar{y}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

The complement of a product is the sum of the complements, and the complement of a sum is the product of the complements. It works for any number of variables: $\overline{ABC} = \bar{A} + \bar{B} + \bar{C}$.

Careful: break the bar *and* swap the operator. $\overline{AB} = \bar{A} + \bar{B}$, never $\bar{A} \cdot \bar{B}$.

In circuits, a NAND is an OR with inverted inputs, and a NOR is an AND with inverted inputs:



$NAND = OR$ with inverted inputs: $\overline{AB} = \bar{A} + \bar{B}$



$NOR = AND$ with inverted inputs: $\overline{A + B} = \bar{A}\bar{B}$

Problem 4

- Prove 15b, $\overline{x + y} = \bar{x} \cdot \bar{y}$: build the truth table for both sides and compare the two columns.
- Rewrite $(A + \bar{B})C$ so that no bar covers more than one letter. Name the theorem at each step.
- Draw an AND gate with an inverter on each input and write down what it computes. Which single gate from section 1 has that truth table? Do the same for an OR gate with an inverter on each input.
- Start from $F = AB + \bar{B}\bar{C}$. Use DeMorgan twice to write the complement $\bar{F}$ as a product of two sums.

5 Algebraic proof and minimisation

Minimising means fewer terms and fewer literals, so fewer gates. Useful moves:

- Multiply out, then drop $x\bar{x} = 0$ (8a) and collapse $y + \bar{y} = 1$ (8b).
- Factor with the distributive law to expose such pairs.
- Duplicate a term ($x + x = x$, 7b) so it can combine with two different neighbours.
- To prove an identity, reduce both sides to the same expression.

The two theorems you will use most: 14a, $xy + \bar{x}\bar{y} = x$, and 16a, $x + \bar{x}y = x + y$.

Problem 5

- Prove the identity $(x_1 + x_3)(\bar{x}_1 + \bar{x}_3) = x_1\bar{x}_3 + \bar{x}_1x_3$.
- What familiar two-input function is this?
- Prove the identity by reducing each side separately: $x_1\bar{x}_3 + \bar{x}_2\bar{x}_3 + x_1x_3 + \bar{x}_2x_3 = \bar{x}_1\bar{x}_2 + x_1x_2 + \bar{x}_1x_2$
- Minimise $f(x_1, x_2) = x_1x_2 + \bar{x}_1\bar{x}_2 + \bar{x}_1x_2$, then check your answer with a truth table.

6 From a truth table to an expression: SOP and POS

Sum of products (SOP). One product per row where the output is 1. A variable is plain if it is 1 in that row, complemented if it is 0. OR the products together.

Product of sums (POS). One sum per row where the output is 0. A variable is plain if it is 0 in that row, complemented if it is 1. AND the sums together.

A	B	F	SOP term	POS term
0	0	1	$\overline{A}\overline{B}$	
0	1	0		$A + \overline{B}$
1	0	0		$\overline{A} + B$
1	1	1	AB	

$$F = \overline{A}\overline{B} + AB = (A + \overline{B})(\overline{A} + B), \text{ which is XNOR.}$$

Careful: the literal rules are opposite. A product term must be 1 on its own row; a sum term must be 0 on its own row.

Inverse logic. Write whichever list is shorter. This table from the slides has only two 0s:

ABC	000	001	010	011	100	101	110	111
F	1	0	1	1	1	0	1	1

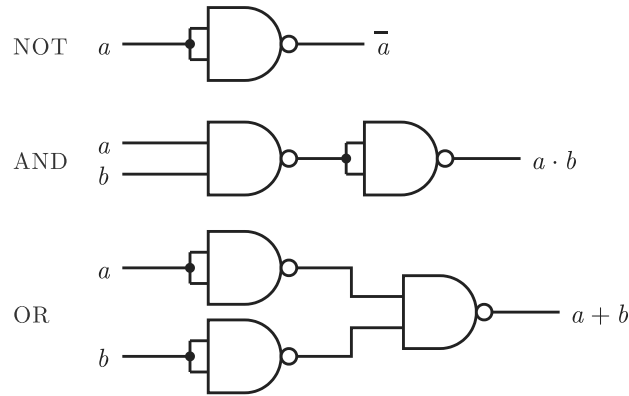
POS: $F = (A + B + \overline{C})(\overline{A} + B + \overline{C}) = B + \overline{C}$ by 14b. Or through the complement: $\overline{F} = \overline{A}\overline{B}C + A\overline{B}C = \overline{B}C$, so $F = B + \overline{C}$.

Problem 6

- For XOR (0 1 1 0 down the rows 00, 01, 10, 11), write the SOP and the POS. Which gates does the SOP need?
- Write the full SOP of $F = AB + B'C'$ from the table you made in Problem 2a (one product per 1), and simplify it back to $AB + B'C'$.
- A three-input function is 0 only on rows 011 and 111. Write it using inverse logic and simplify. What single gate is it?

7 NAND and NOR

NAND alone can make NOT, AND and OR, so it can make any function: it is *universal*. So is NOR.



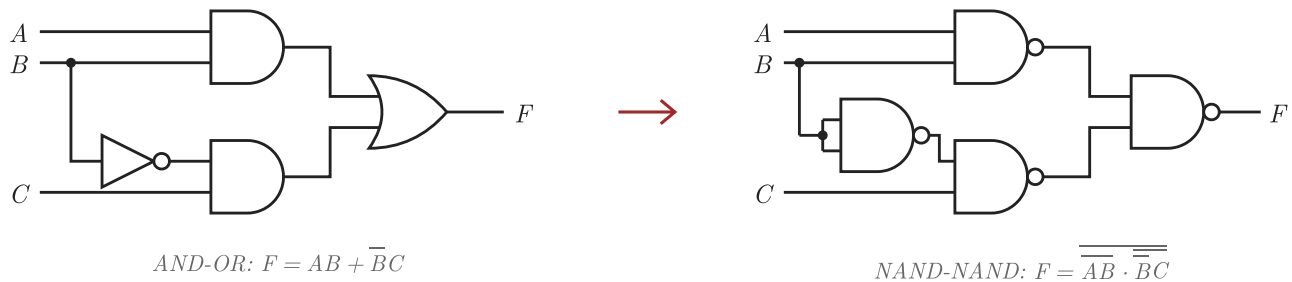
Each construction is one line of algebra. NOT: tie the two inputs, and since $a \cdot a = a$ by 7a the gate gives $\bar{a}$. AND: the first NAND gives $\overline{ab}$ and the second one, inputs tied, inverts it back, $\overline{\overline{ab}} = ab$ by 9. OR: invert both inputs first, then NAND them, $\overline{\bar{a} \cdot \bar{b}} = a + b$ by 15a and 9.

In CMOS a 2-input NAND or NOR takes four transistors, while AND and OR need an extra inverter after them: six transistors and one more gate delay. So real circuits are built from inverting gates.

SOP to NAND-NAND, from the formula. Take $F = AB + \bar{B}C$ and complement it twice, which by 9 changes nothing: $F = \overline{\overline{AB + \bar{B}C}}$. The inner complement sits over a sum, so 15b splits it into a product, $\overline{AB + \bar{B}C} = \overline{AB} \cdot \overline{\bar{B}C}$, and therefore

$$F = \overline{\overline{AB} \cdot \overline{\bar{B}C}}$$

Now read the complements as gates: $\overline{AB}$ is a NAND of A and B , $\overline{\bar{B}C}$ is a NAND of $\bar{B}$ and C , and the outer complement of their product is a third NAND fed by those two. In pictures this is the two-bubble trick: put two complements on each wire between the levels, give the first to the AND, which becomes a NAND, and the second to the OR, which becomes a NAND by DeMorgan.



The rule that comes out of it is short: in a two-level AND-OR circuit, replace every gate with a NAND and leave the wiring alone. A product of sums goes the same way: $(A + B)(\bar{B} + C) = \overline{\overline{A + B} \cdot \overline{\bar{B} + C}}$ by 9 and 15a, and every complement of a sum is a NOR, so an OR-AND circuit becomes all NORs. Counting gates for $F = AB + \bar{B}C$: two NANDs for the products, one for the sum, and one with its inputs tied to make $\bar{B}$, which is four 2-input NANDs, exactly one 74LS00.

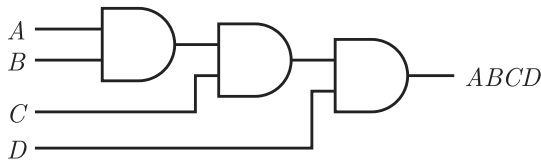
Careful: the rule assumes every input of the OR comes from an AND. A literal that feeds the OR directly has no gate to invert it, so it must enter the output NAND complemented: $AB + C = \overline{\overline{AB} \cdot \bar{C}}$.

Problem 7

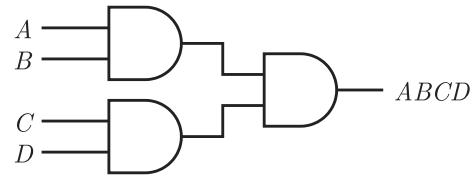
- Build XOR from NAND gates only. Four are enough. (Hint: start from $a\bar{b} + \bar{a}b$ and apply DeMorgan.)
- Show that NOR is also universal by building NOT, OR and AND from NOR gates.
- Convert the POS circuit $F = (A + B)(\bar{B} + C)$ to NOR gates only. How many 2-input NORs does it take?

8 Gates with more than two inputs

Chaining is just regrouping: $A + B + C = (A + B) + C$ by 11b, and $ABCD = ((AB)C)D = (AB)(CD)$ by 11a. Every grouping is the same function, so you may combine the inputs two at a time in any order. An n -input gate takes $n - 1$ two-input gates whichever shape you pick, but the shape sets the delay: in the tree $(AB)(CD)$ every input is two gates from the output, while in the chain $((AB)C)D$ the first two inputs are three gates away.



chain: 3 gates; A and B pass through 3 of them



tree: 3 gates; every input passes through 2

Careful: NAND and NOR do not regroup. Chaining two NANDs gives $\overline{\overline{AB} \cdot C} = AB + \bar{C}$ by 15a and 9, while a 3-input NAND is $\overline{ABC} = \bar{A} + \bar{B} + \bar{C}$. At $A = B = 0, C = 1$ the first is 0 and the second is 1. For three inputs write $\overline{ABC} = \overline{(AB)C}$: an AND first, then a NAND.

Solutions

Problem 1

- With primes instead of bars, $\bar{A}B + A\bar{B}$ is $A'B + AB'$. It is 1 when $A = 0$ and $B = 1$, and when $A = 1$ and $B = 0$, that is, exactly when the inputs differ. That is XOR.
- The order is NOT, then AND, then OR, so $\bar{x}y + z$ means $((\bar{x}) \cdot y) + z$.
- Not the same. Take $x = 1, y = 0, z = 0$: the first is $1 + 0 = 1$, the second is $(1 + 0) \cdot 0 = 0$. To show two functions differ, one row is enough; to show they are equal, every row has to agree.
- Not the same. At $x = 1, y = 0$ the first is $0 \cdot 1 = 0$ and the second is $\bar{0} = 1$. The correct expansion is DeMorgan, 15a: $\overline{xy} = \bar{x} + \bar{y}$.

Problem 2

- AB is 1 on the rows 110 and 111. $\bar{B}\bar{C}$ is 1 where B and C are both 0, rows 000 and 100. The OR of those two columns is 1 on all four of those rows.

A	B	C	AB	$\overline{BC}$	F
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	0	1

- b. $A + B$ is 0 on rows 000 and 001, so G is 0 there whatever the second factor does. $\overline{A} + C$ is 0 when A is 1 and C is 0, rows 100 and 110. The product is 1 only where both columns are 1.

A	B	C	$A + B$	$\overline{A} + C$	G
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	0	0
1	1	1	1	1	1

- c. A three-variable identity needs $2^3 = 8$ rows; a six-variable one needs $2^6 = 64$ rows.

Problem 3

- a. $x(x + y) = x$ directly by absorption, 13b. The long way: $x(x + y) = xx + xy$ by 12a, $= x + xy$ by 7a, $= x$ by 13a.
- b. $x + \overline{xy} = x + y$ by 16a.
- c. Swap $\cdot$ with $+$ and 0 with 1: the dual of $x(y + z) = xy + xz$ is $x + yz = (x + y)(x + z)$, which is 12b. This one has no counterpart in ordinary arithmetic, where addition does not distribute over multiplication.
- d. The first two terms differ only in y , so 14a merges them: $xy + x\overline{y} = x$. What is left is $x + \overline{xy}$, which is $x + y$ by 16a.
- e. Substitute into the definition. $A \oplus 1 = \overline{A} \cdot 1 + A \cdot \overline{1}$; by 4b the complement of 1 is 0, so this is $\overline{A} \cdot 1 + A \cdot 0$, which is $\overline{A} + 0$ by 6a and 5a, and $\overline{A}$ by 6b. For the second one, $A \oplus \overline{A} = \overline{A} \cdot \overline{A} + A \cdot \overline{\overline{A}}$; by 9 the double complement is A , so this is $\overline{A}\overline{A} + AA$, which is $\overline{A} + A$ by 7a, and 1 by 10b and 8b.

Problem 4

- a. Build a column for each side. Both read 1, 0, 0, 0 down the four rows, so the identity holds.

x	y	$x + y$	$\overline{x + y}$	$\overline{x}$	$\overline{y}$	$\overline{xy}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

- b. Start at the outermost bar, which covers a product, so 15a turns it into a sum: $\overline{(A + B)C} = \overline{A + B} + \overline{C}$. The first part still has a bar over a sum, so 15b turns that into a product: $\overline{A + B} = \overline{A} \cdot \overline{B}$. Finally $\overline{\overline{B}} = B$ by 9, so the answer is $\overline{A}B + \overline{C}$. The operator swapped at every step.
- c. The AND with an inverter on each input computes $\overline{A}\overline{B}$, which is $\overline{A + B}$ by 15b: that is the NOR truth table. The OR with inverted inputs computes $\overline{A} + \overline{B}$, which is $\overline{AB}$ by 15a: the NAND truth table.
- d. The bar covers a sum of two products, so 15b splits it into a product of two bars: $\overline{F} = \overline{AB} \cdot \overline{BC}$. Then 15a opens each factor, $\overline{AB} = \overline{A} + \overline{B}$ and $\overline{BC} = \overline{B} + \overline{C}$, and 9 removes the double bars, leaving $\overline{F} = (\overline{A} + \overline{B})(\overline{B} + \overline{C})$, a product of two sums.

Problem 5

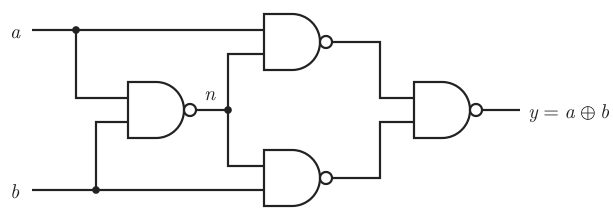
- a. Expand with 12a: $(x_1 + x_3)(\overline{x_1} + \overline{x_3}) = (x_1 + x_3)\overline{x_1} + (x_1 + x_3)\overline{x_3}$, which multiplies out to $x_1\overline{x_1} + x_3\overline{x_1} + x_1\overline{x_3} + x_3\overline{x_3}$. The first and last terms are 0 by 8a, and 6b drops them, leaving $x_1\overline{x_3} + \overline{x_1}x_3$.
- b. XOR: it is 1 exactly when the two inputs differ.
- c. Left side: group the terms in pairs, $x_1(\overline{x_3} + x_3) + \overline{x_1}(\overline{x_3} + x_3)$ by 12a; each bracket is 1 by 8b, so 6a gives $x_1 + \overline{x_1}$. Right side: $\overline{x_1}x_2 + x_1(\overline{x_2} + x_2) = \overline{x_1}x_2 + x_1$ by 8b and 6a, which is $x_1 + \overline{x_1}x_2$ by 10b, and $x_1 + \overline{x_2}$ by 16a. Both sides equal $x_1 + \overline{x_2}$.
- d. Two routes give the same answer. With 14a: $x_1x_2 + \overline{x_1}x_2 = x_2$, and what is left is $x_2 + \overline{x_1}x_2$, which is $x_2 + \overline{x_1}$ by 16a. Or duplicate a term with 7b: $f = x_1x_2 + \overline{x_1}x_2 + \overline{x_1}x_2 + \overline{x_1}x_2$, then factor the first pair on x_2 and the second on $\overline{x_1}$: $(x_1 + \overline{x_1})x_2 + \overline{x_1}(\overline{x_2} + x_2) = x_2 + \overline{x_1}$ by 8b and 6a. Check: f is 1 on the rows 00, 01 and 11 and 0 on 10, and $x_2 + \overline{x_1}$ is 0 only when $x_1 = 1, x_2 = 0$.

Problem 6

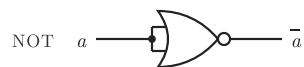
- a. The 1-rows are 01 and 10, so the SOP is $\overline{A}B + A\overline{B}$. The 0-rows are 00 and 11, so the POS is $(A + B)(\overline{A} + \overline{B})$. The SOP circuit needs two inverters, two 2-input ANDs and one OR. These two expressions are the two sides of the identity in Problem 5a.
- b. The 1-rows are 000, 100, 110 and 111, one product each: $F = \overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C} + AB\overline{C} + ABC$. The first pair differs only in A , so 14a merges it into $\overline{B}\overline{C}$; the second pair differs only in C , giving AB . So $F = AB + \overline{B}\overline{C}$. The full SOP is easy to write but long, and simplifying shrinks the circuit a lot.
- c. Only two rows are 0, so work with the complement: $\overline{F} = \overline{A}BC + ABC = BC$ by 14a, and therefore $F = \overline{BC} = \overline{B} + \overline{C}$ by 15a. That is a single NAND of B and C ; A does not matter. The POS route gives the same thing: $(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + \overline{C}) = \overline{B} + \overline{C}$ by 14b.

Problem 7

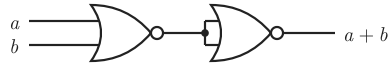
- a. Name the first gate's output $n = \overline{ab}$. The next two gates give $\overline{an}$ and $\overline{bn}$, and the last one combines them, so $F = \overline{\overline{an} \cdot \overline{bn}}$. By 15a and 9 that opens into $F = an + bn$. Now $n = \overline{a} + \overline{b}$ by 15a, so $an = a(\overline{a} + \overline{b}) = a\overline{a} + a\overline{b} = a\overline{b}$ by 12a, 8a and 6b, and in the same way $bn = \overline{a}b$. So $F = a\overline{b} + \overline{a}b$, which is 1 exactly when the inputs differ: XOR, from four NAND gates.
- b. NOR mirrors NAND. NOT: tie the inputs, and $\overline{a + a} = \overline{a}$ by 7b. OR: a NOR followed by that inverter, $\overline{\overline{a + b}} = a + b$ by 9. AND: invert both inputs, then NOR them, $\overline{\overline{a} + \overline{b}} = ab$ by 15b and 9.
- c. Complement twice, which changes nothing by 9, then split the inner bar with 15a: $F = (A + B)(\overline{B} + C) = \overline{\overline{A + B} + \overline{\overline{B} + C}}$. Every bar over a sum is a NOR, so this is one NOR for $A + B$, one for $\overline{B} + C$, and a third combining them. One more NOR with its inputs tied makes $\overline{B}$: four 2-input NORs in total, exactly one 74LS02.



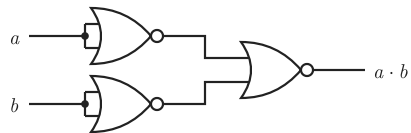
(a) XOR from four NANDs



NOT



OR



AND

(b) NOT, OR and AND from NORs